

UNIT-2

Data Preprocessing

Study Guide

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- Introduction to Data Preprocessing
- Reading, Selecting
- Filtering Data – Filtering Missing Values
- Manipulating and Sorting Data
- Grouping, Rearranging Data
- Ranking Data

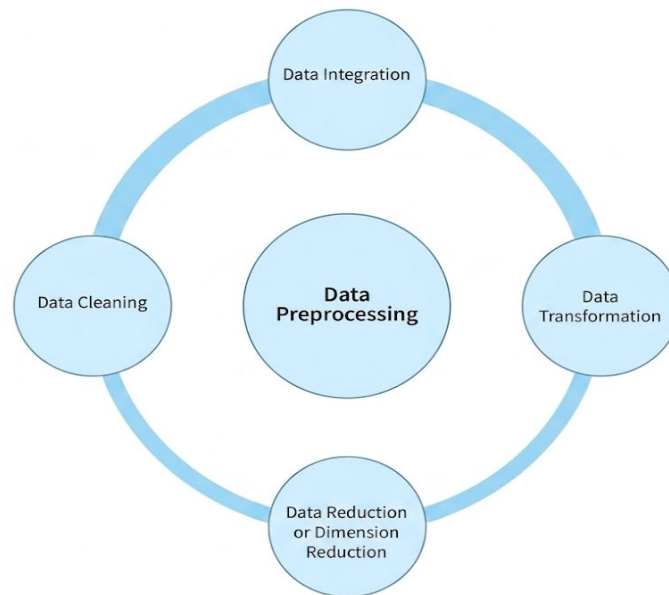
Data Pre-processing

- Data Preprocessing is an essential stage in the Data Preparation phase of a Data Science development lifecycle.
- The process of converting raw data into a format that can be processed more efficiently and accurately.
- Data preprocessing resolves issues such as inconsistent formatting, human errors, and incomplete data.
- Data preprocessing makes datasets more complete and efficient to carry out data analysis.
- In other words, data preprocessing is transforming data into a form that computers can easily work on. It makes data analysis or visualization easier and increases the accuracy and speed of the machine learning algorithms that train on the data.

Factors affecting Data Preprocessing:

- ❖ **Accuracy:** Accuracy means that the information is correct.
- ❖ **Consistency:** The data should be consistent. Inconsistent data may result in different answers to the same question.
- ❖ **Completeness:** The dataset shouldn't have incomplete fields or missing fields.
- ❖ **Validity:** A dataset is considered valid if the data samples appear in the correct format, are within a specified range, and are of the right type. **Timeliness** - Whether data is updated regularly and on a timely basis or not.
- ❖ **Trustable** - Whether data is coming from trustworthy sources or not.
- ❖ **Interpretability** - Raw data is generally unusable, and Data Pre-processing converts raw data into an interpretable format.

Key Steps in Data Pre-processing



➤ Data Cleaning

Data Cleaning uses methods to handle incorrect, incomplete, inconsistent, or missing values. Some of the techniques for Data Cleaning include -

- **Handling Missing values**

Input data can contain missing or **NULL** values, which must be handled before applying any Machine Learning or Data Mining techniques.

Missing values can be handled by many techniques, such as removing rows/columns containing **NULL** values and imputing **NULL** values using mean, mode, regression, etc.

- **De-noising**

- De-noising is a process of removing noise from the data. Noisy data is meaningless data that is not interpretable or understandable by machines or humans. It can occur due to data entry errors, faulty data collection, etc.
- De-noising can be performed by applying many techniques, such as binning the features, using regression to smoothen the features to reduce noise, clustering to detect the outliers, etc.

➤ Data Integration

Data Integration can be defined as combining data from multiple sources. A few of the issues to be considered during Data Integration include the following -

- **Entity Identification Problem** - It can be defined as identifying objects/features from multiple databases that correspond to the same entity. For example, in database A_customer_id_ and in database B_customer_number_ belong to the same entity.
- **Schema Integration** - It is used to merge two or more database schema/metadata into a single schema. It essentially takes two or more schema as input and determines a mapping between them. For example, entity type CUSTOMER in one schema may have CLIENT in another schema.
- **Detecting and Resolving Data Value Concepts** - The data can be stored in various ways in different databases, and it needs to be taken care of while integrating them into a single dataset. For example, dates can be stored in various formats such as **DD/MM/YYYY**, **YYYY/MM/DD**, or **MM/DD/YYYY**, etc.

➤ Data Reduction

Data Reduction is used to reduce the volume or size of the input data. Its main objective is to reduce storage and analysis costs and improve storage efficiency. A few of the popular techniques to perform Data Reduction include -

- **Dimensionality Reduction** - It is the process of reducing the number of features in the input dataset. It can be performed in various ways, such as selecting features with the highest importance, Principal Component Analysis (PCA), etc.
- **Numerosity Reduction** - In this method, various techniques can be applied to reduce the volume of data by choosing alternative smaller representations of the data. For example, a variable can be approximated by a regression model, and instead of storing the entire variable, we can store the regression model to approximate it.
- **Data Compression** - In this method, data is compressed. Data Compression can be lossless or lossy depending on whether the information is lost or not during compression.

➤ Data Transformation

Data Transformation is a process of converting data into a format that helps in building efficient ML models and deriving better insights. Techniques for Data Transformation include -

- **Smoothing** - Data Smoothing is used to remove noise in the dataset, and it helps identify important features and detect patterns. Therefore, it can help in predicting trends or future events.
- **Aggregation** - Data Aggregation is the process of transforming large volumes of data into an organized and summarized format that is more understandable and comprehensive. For example, a company may look at monthly sales data of a product instead of raw sales data to understand its performance better and forecast future sales.
- **Discretization** - Data Discretization is a process of converting numerical or continuous variables into a set of intervals/bins. This makes data easier to analyze. For example, the age features can be converted into various intervals such as (0-10, 11-20, ..) or (child, young, ...).
- **Normalization** - Data Normalization is a process of converting a numeric variable into a specified range such as [-1,1], [0,1], etc. A few of the most common approaches to performing normalization are Min-Max Normalization, Data Standardization or Data Scaling, etc.

❖ **Applications of Data Pre-processing**

Data Pre-processing is important in the early stages of a Machine Learning and AI application development lifecycle.

Improved Accuracy of ML Models -

Reduced Costs

Visualization

❖ Reading Data

Reading data is a fundamental step in the data science pipeline, setting the stage for analysis, modeling, and interpretation. This process involves acquiring, understanding, and preparing data from various sources. A comprehensive understanding of this topic covers several key areas, including the sources of data, methods of data ingestion, challenges faced during data acquisition, and best practices to ensure data quality and reliability.

Introduction to Reading Data

Reading data refers to the process of importing and loading data from various sources into a data science environment. This step is crucial as it forms the foundation upon which all subsequent analysis is built. The ability to efficiently and accurately read data is essential for successful data science projects.

Sources of Data

- **Structured Data**

Databases: Relational databases (e.g., MySQL, PostgreSQL) and NoSQL databases (e.g., MongoDB).

Spreadsheets: CSV, Excel files.

- **Unstructured Data**

Text Files: Plain text, JSON, XML.

Multimedia: Images, audio, video files.

- **Semi-Structured Data**

Logs: Server logs, application logs.

- **Web Data:** HTML pages, web scraping outputs.

APIs and Streaming Data

- **APIs:** RESTful APIs, SOAP APIs for accessing web services.

- **Streaming:** Real-time data from sources like Kafka, MQTT.

❖ Selecting Data

Selecting data in data science involves the process of identifying and extracting relevant subsets of data from larger datasets for analysis, modeling, or visualization. This step is crucial for focusing on specific questions or objectives and ensuring that the analysis is based on the most pertinent information.

Key Concepts in Selecting Data

Relevance: Choosing data that is directly related to the problem or question at hand.

Validity: Ensuring that the selected data is accurate, reliable, and representative of the underlying phenomena.

Completeness: Including all necessary variables or attributes required for the analysis.

Quality: Selecting data that meets certain quality standards, free from errors, inconsistencies, or biases.

Dimensionality: Considering the number of features or dimensions in the dataset and selecting only those that contribute meaningfully to the analysis.

Balance: Ensuring that the selected data is representative across different classes or categories, especially in classification tasks.

Size: Balancing the trade-off between data size and computational resources available for analysis.

❖ Filtering Data

Filtering data in data science involves the process of selecting specific subsets of data based on certain conditions or criteria. This step is essential for focusing the analysis on relevant information, removing noise, and preparing the data for further exploration, modeling, or visualization.

- **Filter by Conditions**
 - **Numerical Filtering**
 - **Text Filtering:**
 - **Categorical Filtering**
- **Filter by Rows or Columns**
 - Row Filtering:
 - Column Filtering:
- **Filter by Data Quality**
- **Filter by Time**

Filtering missing values, also known as handling missing data, is a crucial step in data preprocessing in data science.

❖ Manipulating data

Manipulating data is a fundamental part of the data science workflow. It involves transforming raw data into a format that is suitable for analysis. This process includes cleaning, reshaping, aggregating, and merging data, among other operations.

➤ Key Concepts in Data Manipulation

- **Data Cleaning:** Removing or correcting errors and inconsistencies in the data.
- **Data Transformation:** Changing the format or structure of the data.
- **Data Aggregation:** Summarizing data by grouping and calculating aggregate statistics.
- **Data Merging:** Combining data from different sources.
- **Feature Engineering:** Creating new features from existing data to improve model performance.

➤ Methods of Data Manipulation

- **Data Cleaning**
- **Handling Missing Values:**
- **Remove:** Drop rows or columns with missing values.
- **Impute:** Fill missing values with mean, median, mode, or other values.
- **Interpolate:** Use interpolation methods for filling missing values in time series data.
- **Removing Duplicates:**
- **Drop Duplicates:** Remove duplicate rows using functions like `drop_duplicates()` in pandas.
- **Correcting Errors:**
- **Data Validation:** Use domain knowledge to correct errors in the data.
- **Regex:** Use regular expressions to clean text data.
- **Data Transformation**
- **Normalization/Standardization:**
- **Scaling:** Scale data to a standard range (e.g., 0 to 1) or to have zero mean and unit variance.
- **Encoding Categorical Variables:**
- **Label Encoding:** Convert categorical labels to numerical values.
- **One-Hot Encoding**
- **Data Aggregation**

❖ Sorting

Sorting data is a fundamental operation in data science that involves arranging data in a particular order, typically ascending or descending, based on one or more attributes. Sorting is crucial for data analysis, visualization, and preparation for further operations. **Key Concepts in Sorting Data**

- **Ordering:** Arranging data in ascending or descending order based on the values of one or more columns.
- **Stability:** Maintaining the relative order of records with equal keys (values) after sorting.
- **Multi-Level Sorting:** Sorting data based on multiple columns, where the order of precedence is specified.

❖ **Methods for Sorting Data**

- **Single-Column Sorting:** Sorting data based on the values of a single column.
- **Multi-Column Sorting:** Sorting data based on the values of multiple columns, with a defined order of precedence.

Sorting in Different Environments Python (using pandas)

Pandas is a powerful Python library for data manipulation and analysis.

Single-Column Sorting

```
import pandas as pd # Sample DataFrame data = {  
'id': [1, 2, 3, 4, 5],  
'name': ['Alice', 'Bob', 'Charlie', 'David', 'Edward'], 'age': [25, 35, 45, 20, 30]  
}  
df = pd.DataFrame(data)  
# Sort by age in ascending order sorted_df = df.sort_values(by='age') print(sorted_df)
```

R (using dplyr)

dplyr is a powerful R package for data manipulation. Here's how to sort data using dplyr:

Single-Column Sorting

```
library(dplyr)  
# Sample data frame df <- data.frame(id = c(1, 2, 3, 4, 5), name = c('Alice', 'Bob', 'Charlie', 'David',  
'Edward'), age = c(25, 35, 45, 20, 30))
```

❖ Grouping

Grouping data is a powerful technique in data science used to aggregate and summarize data, making it easier to analyze and derive insights from large datasets. Grouping involves splitting the data into subsets based on one or more criteria and then applying aggregate functions to each subset. This method is particularly useful for identifying patterns, trends, and relationships within the data.

❖ Key Concepts in Grouping Data

- **Grouping:** Splitting the dataset into subsets based on the values of one or more columns.
- **Aggregation:** Applying functions to each group to compute summary statistics such as mean, sum, count, max, and min.
- **Transformation:** Applying functions to each group to transform the data.
- **Filtering:** Selecting groups based on aggregate criteria.
- **Grouping in Different Environments Python (using pandas)**
- **Pandas is a powerful library for data manipulation in Python. Here's how to perform grouping using pandas:**

- **Grouping by a Single Column**

```
import pandas as pd # Sample DataFrame
```

```
data = { 'id': [1, 2, 3, 4, 5],
```

```
'name': ['Alice', 'Bob', 'Charlie', 'David', 'Edward'], 'age': [25, 35, 45, 20, 30], 'sales': [100, 200, 300, 400, 500]}
```

```
df = pd.DataFrame(data)
```

```
# Group by 'name' and calculate the sum of 'sales' grouped_
```

```
df = df.groupby('name')['sales'].sum().reset_index() print(grouped_df)
```

- **R (using dplyr)**

dplyr is a powerful R package for data manipulation. Here's how to perform grouping using dplyr:

Grouping by a Single Column

```
library(dplyr)
```

```
# Sample data frame
```

```
df <- data.frame( id = c(1, 2, 3, 4, 5), name = c('Alice', 'Bob', 'Charlie', 'David', 'Edward'), age = c(25, 35, 45, 20, 30), sales = c(100, 200, 300, 400, 500) )
```

```
# Group by 'name' and calculate the sum of 'sales' grouped_df <- df %>% group_by(name) %>%
```

```
summarise(sales_sum = sum(sales)) print(grouped_df)
```

❖ Rearranging

Rearranging data in data science involves organizing, transforming, and restructuring raw data into a format that is more suitable for analysis or modeling. This process, often referred to as data manipulation or data wrangling, is a fundamental step in the data science workflow. common techniques and tools used for rearranging data:

- **Data Cleaning**
- **Reshaping Data**
- **Subset Selection**
- **Sorting**
- **Joining and Merging**
- **Grouping and Aggregating**
- **Feature Engineering**
- **Time Series Manipulation**

❖ **Ranking**

Ranking data in data science involves assigning a numerical value to each item in a dataset based on a specific criterion or set of criteria. Ranking is commonly used in various analytical tasks such as identifying top performers, prioritizing items, or evaluating the relative importance of different elements in a dataset. techniques for ranking data:

- **Simple Ranking:** Assigning ranks to items based on a single criterion, such as their numerical value or performance score.
- **Percentile Ranking:** Dividing the data into equal-sized groups (percentiles) and assigning ranks based on the percentile each item falls into.
- **Rank Aggregation:** Combining rankings from multiple sources or criteria to generate a consensus ranking.
- **Tie-breaking:** Handling ties that occur when multiple items have the same rank.
- **Ranking with Weighted Criteria:** Assigning ranks based on multiple criteria with different weights or importance levels.
- **Ranking Time Series Data:** Ranking data over time to identify trends, fluctuations, or changes in rank positions.
- **Dynamic Ranking:** Updating ranks in real-time as new data becomes available.

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